HRSF: A Hierarchical Residual Distribution Framework for Remote Sensing Image Fusion Under Spatiotemporal Constraints

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***Abstract*—Spatiotemporal fusion integrates multisource remote sensing data to produce imagery that simultaneously possesses high spatial and temporal resolution. However, existing spatiotemporal fusion approaches still face limitations in reconstructing abrupt land-cover changes while preserving spatial structures, often leading to structural distortions and error propagation. To address this limitation, the shared change patterns among homogeneous pixels are explicitly modeled to improve structural reconstruction under abrupt changes, and a hierarchical residual distribution spatiotemporal fusion framework (HRSF) framework is further proposed. The framework progressively integrates temporal consistency with spatial structural constraints through hierarchical residual propagation, imposing both explicit and implicit constraints on residual propagation, thereby enabling unified modeling of temporal evolution and spatial structure. Furthermore, building upon the pixel-level residual redistribution, a dual-ranking consistency strategy is introduced to ensure stable and physically consistent similar-pixel selection. HRSF is evaluated against eight state-of-the-art spatiotemporal fusion methods across four distinct landscapes. Quantitative results demonstrate that HRSF consistently achieves superior predictive accuracy across different datasets and sensor configurations. Overall, HRSF effectively balances spatial structural stability and spectral consistency, enabling high-fidelity spatiotemporal fusion for dynamic environmental monitoring. The Python code of HRSF and the experimental dataset are available at: https://github.com/paboooo/HRSF**

***Index Terms*—Hierarchical residual model, image reconstruction, remote sensing, spatiotemporal fusion, time series.**

NOMENCLATURE

TR	Temporal residual estimation.
OR	Object-constrained residual compensation.
CR	Context-aware residual compensation.
CF	Confidence-guided adaptive fusion.
$F_{T_1}(x_{ij}, y_{ij}, b)$	jth fine-resolution pixel within the ith coarse-resolution pixel in band b at T_1.
$C_{T_1}(x_i, y_i, b)$/ $C_{T_2}(x_i, y_i, b)$	ith coarse-resolution pixel in band b at T_1/T_2.
$F^{BIC}_{T_2}(x_{ij}, y_{ij}, b)$	Coarse image at T_2 upsampled using bicubic interpolation.
$R^{TR}(x_{ij}, y_{ij}, b)$/ $R^{OR}(x_{ij}, y_{ij}, b)$	Fine-resolution residual obtained by subtracting the downsampled $F^{TR}_{T_2}(x_i, y_i, b)/F^{OR}_{T_2}(x_i, y_i, b)$ from the coarse-resolution pixel at T_1.
$\Delta F^{TR}(n, b)$	Temporal change of the nth fine-resolution pixel belonging to class c.
$\Delta F^{OR}(x_{ij}, y_{ij}, b)$/ $\Delta F^{CR}(x_{ij}, y_{ij}, b)$	Redistributed residual obtained by applying the OR/CR to $R^{TR}(x_{ij}, y_{ij}, b)/R^{OR}(x_{ij}, y_{ij}, b)$.
w	Size of moving window for spectral unmixing.
n	Fine-resolution pixel of class c.
N	Number of fine-resolution pixels belonging to class c within the corresponding coarse-resolution pixel.
s	Spatial resolution ratio between the fine and coarse-resolution images.
n_b	Number of spectral bands.
$r\%/r_s$	Percentage and number of reference samples used for residual estimation.
k	Neighboring fine-resolution pixel within the same coarse pixel as the target pixel.
k_s	Number of selected similar pixels.
r_t	Ranking-consistency threshold.
w_s	Size of the moving window when searching for similar pixels.
$F^{TR}_{T_2}(x_{ij}, y_{ij}, b)$/ $F^{OR}_{T_2}(x_{ij}, y_{ij}, b)$/ $F^{CR}_{T_2}(x_{ij}, y_{ij}, b)$/ $F^{CF}_{T_2}(x_{ij}, y_{ij}, b)$	Prediction of the fine-resolution pixel after TR/OR/CR/CF.

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I. Introduction

REMOTE sensing satellite imagery has been widely used for environmental monitoring [1], urban planning [2], disaster management [3], and resource exploration [4]. However, due to sensor limitations and environmental conditions [5], satellite observations generally cannot simultaneously provide high spatial and temporal resolution [6], [7], [8]. For instance, Landsat TM/ETM+ and SPOT HRV provide imagery at spatial resolutions of 10–30 m but with relatively long revisit intervals (Landsat TM/ETM+: 16 days; SPOT HRV: 26 days), which, together with frequent cloud contamination, limit their effectiveness in applications requiring frequent observations. In contrast, the Sentinel-3 Ocean and Land color Instrument (OLCI) provides daily observations but at a relatively coarse spatial resolution of 300 m. Consequently, obtaining imagery that simultaneously achieves high spatial and temporal resolution remains a significant challenge. To address this limitation, spatiotemporal fusion techniques have been widely developed, aiming to enhance both spatial and temporal resolution by integrating complementary information from coarse- and fine-resolution observations [9], [10], [11].

Existing spatiotemporal fusion methods are broadly categorized into nonlearning-based and learning-based methods. Nonlearning-based methods can be further divided into unmixing-based, weighted function-based, and hybrid methods [11], [12]. The multisensor multiresolution technique (MMT) [13] was among the first methods to introduce spectral unmixing into spatiotemporal fusion. The linear spectral unmixing framework introduced by MMT has inspired several subsequent unmixing-based models, such as STDFA [14], the unmixing-based spatiotemporal reflectance fusion model [15], the geographically weighted spatial unmixing method [16], and spatial unmixing with block removal [17]. These methods further improved the accuracy and robustness of the unmixing-based process. Weight function-based methods achieve spatiotemporal fusion by combining the spectral information of similar neighboring pixels. Representative approaches include the spatial and temporal adaptive reflectance fusion model (STARFM) [18], its enhanced version ESTARFM [3], and the three-step method (Fit-FC method) [19], all of which have significantly improved the prediction accuracy of weighting-based spatiotemporal fusion methods. These developments subsequently led to a series of hybrid approaches, including flexible spatiotemporal data fusion (FSDAF) [20], reliable and adaptive spatiotemporal data fusion (RASDF) [21], and variational spatiotemporal data fusion (VSDF) [22], which integrate spectral unmixing with weighting strategies to enhance prediction robustness. Recently, the rapid advances in deep learning have led to the emergence of a large number of learning-based spatiotemporal fusion methods. From the perspective of network architecture, these methods can be broadly classified into five categories [23]: convolutional neural networks (CNNs) [24], Transformers [25], generative models [26], sequence models [27], and other emerging architectures, such as graph neural networks [28], dual-branch fusion networks [29], and multilayer perceptrons [30].

In practical applications, acquiring high-quality paired observations remains challenging due to cloud contamination, revisit limitations, and cross-sensor discrepancies [31]. Although recent learning-based spatiotemporal fusion methods have demonstrated strong capabilities in modeling complex spatial and temporal relationships, they generally require large amounts of high-quality training data, leading to high computational costs and limiting their applicability in large-scale operational scenarios [32]. In contrast, nonlearning-based methods remain attractive because they can achieve competitive fusion performance from limited observations without extensive training.

Nevertheless, existing nonlearning-based methods still suffer from inherent limitations. Spectral unmixing-based methods usually rely on simplified mixture assumptions, resulting in block artifacts and limited capability in modeling intraobject temporal variations [33]. Due to spectral discrepancies between sensors and temporal inconsistencies caused by land surface changes, the weight function-based methods may neglect temporal variation [14], [18] or fail to preserve spatial structures [19], [20]. Specifically, existing weighted function-based methods mainly propagate residuals using either object-level constraints or local neighborhood similarity. However, these two strategies exhibit complementary limitations. Object-level approaches effectively preserve structural consistency but often overlook fine-scale local temporal variations. In contrast, pixel-level approaches are more flexible in capturing local changes, yet they may propagate inconsistent residuals in heterogeneous regions because of the lack of explicit structural constraints. Consequently, existing methods still struggle to jointly preserve faithful temporal evolution and stable spatial structures during residual propagation.

To address this challenge, this study proposes a hierarchical residual distribution spatiotemporal fusion framework (HRSF), which operates on a single fine-coarse image pair and one coarse image acquired at the prediction date. Instead of relying on a single residual propagation strategy, HRSF performs residual compensation in a hierarchical manner by progressively introducing object-scale structural constraints and pixel-scale temporal consistency constraints. Specifically, object-constrained residual propagation first establishes globally consistent structural guidance, after which context-aware propagation further refines local residual distribution through the proposed spectral-temporal dual-ranking consistency criterion. In this way, explicit structural constraints and implicit temporal consistency are jointly enforced during hierarchical residual propagation, enabling faithful temporal evolution reconstruction while preserving stable spatial structures. Finally, a change-spatial information index (CSII)-derived confidence weighting strategy adaptively integrates hierarchical residual estimates according to the reliability of temporal information.

The main contributions are summarized as follows.

1) It alleviates structural degradation and biased residual allocation under abrupt land-cover changes and heterogeneous spatial contexts through hierarchical residual propagation, thereby improving the reconstruction of both spatial structures and temporal changes.
2) It improves the reconstruction of newly emerged land-cover types within the same class.

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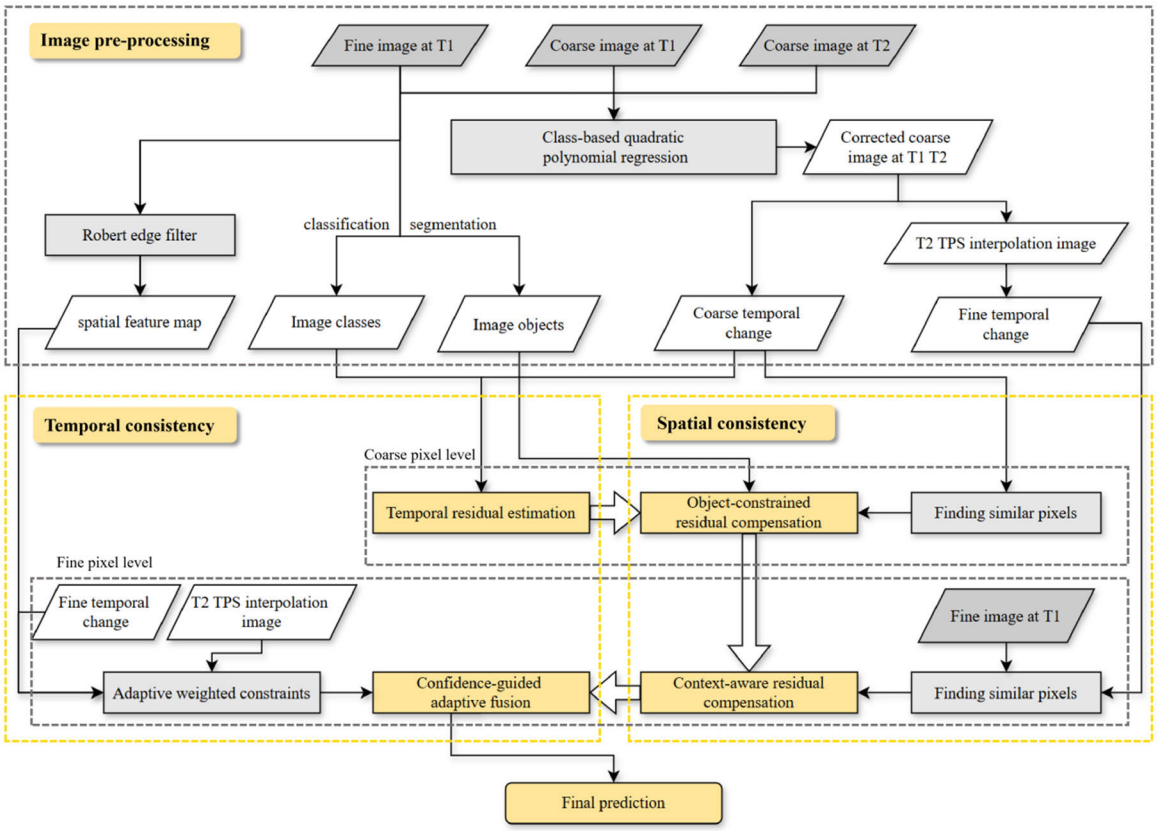


Fig. 1. Flowchart of HRSF.

3) It provides a simple and practical preprocessing strategy to enhance the quality of cross-sensor image pairs.
4) HRSF achieves reliable and robust performance across varying temporal intervals, sensor configurations, and practical application scenarios.

The remainder of this article is organized as follows. Section II describes the methodological components of HRSF. Section III presents the comparative experiments. Section IV presents an application experiment supported by HRSF. Sections V and VI provide further discussion and concluding remarks, respectively.

II. Methodology

The HRSF performs spatiotemporal fusion using one pair of fine-coarse-resolution images acquired at the base time (T_1) and a coarse-resolution image at the prediction time (T_2) to predict the fine-resolution image at T_2. A list of important notations and definitions is given in the Nomenclature for convenience. The overall technical workflow of HRSF is illustrated in Fig. 1.

A. Hierarchical Residual-Consistency Formulation

In multisource spatiotemporal fusion, the target fine-resolution image should inherit the temporal variation observed by the coarse sensor while retaining the spatial structures contained in the base-date fine-resolution image. Let the coarse-scale temporal change be denoted by $\Delta C = C_2 - C_1$. Let $S(\cdot)$ be the fine-to-coarse aggregation operator, $U(\cdot)$ be the coarse-to-fine interpolation operator, and $T(\cdot)$ be the class-level temporal-change estimator. For a current fine-resolution estimate X, a graph-guided residual compensation operator $R_G(\cdot)$ is defined in (1). The complete HRSF mapping is then expressed in (2)

$$R_G(X) = X + P_G\{U[C_2 - S(X)]\} \tag{1}$$

$$\widehat{F}_2 = \gamma \odot U(C_2) + (1-\gamma) \odot \left(R_{G_O} \circ R_{G_C}\right)[F_1 + T(\Delta C)]. \tag{2}$$

The operator P_G denotes normalized residual propagation over graph G in (1). The object-constrained residual estimator and the context-aware residual estimator can be regarded as two graph-specific implementations of this unified propagation operator, where G_O and G_C encode different residual transmission constraints. Accordingly, in (2), R_{G_O} and R_{G_C} represent the object-level and context-level residual compensation operators in Sections II-D and II-E, respectively. The coefficient γ is the *CSII*-derived confidence weight, and the centered composition symbol denotes operator composition. Thus, the object-constrained operator is applied first, after which the context operator acts on the updated prediction.

The composition in (2) establishes an explicit dependency among Sections II-C–II-F: Section II-C provides the initial temporal estimate. Section II-D computes the discrepancy that remains between its aggregated prediction and the observed coarse image at T_2 and redistributes this discrepancy over the object-constrained graph. Section II-E recomputes the

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remaining discrepancy after object-level compensation and propagates only this unexplained component over the finer context graph. Section II-F does not introduce an additional residual correction; instead, it determines the pixelwise confidence placed in the hierarchical residual estimate. HRSF therefore implements a coarse-to-fine relaxation of the homogeneity assumption followed by explicit reliability control.

B. Class-Based Quadratic Polynomial Fitting

Surface reflectance of the same land-cover type should theoretically be consistent on the same date. However, cross-sensor differences (e.g., spectral response, atmospheric conditions, and observation geometry) introduce systematic spectral discrepancies, which degrade fusion accuracy. Although linear regression is commonly used for cross-sensor correction [34], [35], it fails to capture nonlinear reflectance variations and is further limited by spatial heterogeneity. To address this issue, HRSF employs a classwise quadratic regression strategy. The fine-resolution image at T_1 is first partitioned via unsupervised classification, and regression models are estimated separately for each class. The coarse-resolution image at T_2 is then corrected using the corresponding class-specific models, resulting in improved spectral consistency.

C. TR via Spectral Unmixing

Let the coarse-scale temporal change be defined as the difference between the corrected coarse-resolution observations at T_1 and T_2. Under the assumption that pixels belonging to the same land-cover class exhibit comparable temporal behavior, the coarse changes within a local moving window of size $w \times w$ form an overdetermined linear mixture system [3], [14], [20], [33], [35], [36]. Least-squares estimation is employed to recover the class-level fine-resolution temporal changes, represented by $T(\cdot)$ in (2). The estimated class-level temporal changes are then assigned to the fine-resolution pixels of the corresponding land-cover class within each coarse-resolution pixel, resulting in the temporal change of the nth fine-resolution pixel of class c, denoted as $\Delta F^{\mathrm{TR}}(n, b)$

$$F^{\mathrm{TR}}_{\mathrm{T}_2}\left(x_{ij}, y_{ij}, b\right) = F_{\mathrm{T}_1}\left(x_{ij}, y_{ij}, b\right) + \frac{1}{N}\sum_{n=1}^{N}\Delta F^{\mathrm{TR}}(n, b) \quad (3)$$

where N denotes the number of fine-resolution pixels belonging to class c within the coarse pixel at (x_i, y_i), $F^{\mathrm{TR}}_{\mathrm{T}_2}(x_{ij}, y_{ij}, b)$ denotes the initial prediction of the fine-resolution pixel after TR, which is denoted as the initial temporal prediction and serves as the input to the first residual compensation operator in (2).

D. Object-Constrained Residual Compensation

Temporal residual estimation (TR) assumes that land-cover composition is stable between T_1 and T_2. When strong or subobject changes violate this assumption, the aggregated temporal prediction cannot fully match the coarse observation at T_2. In the unified formulation, this unexplained component is measured directly as the difference between C_{T_2} and the aggregated temporal prediction. The object-constrained residual compensation (OR) redistributes this residual while allowing the observed coarse image at T_2 to continuously constrain the correction.

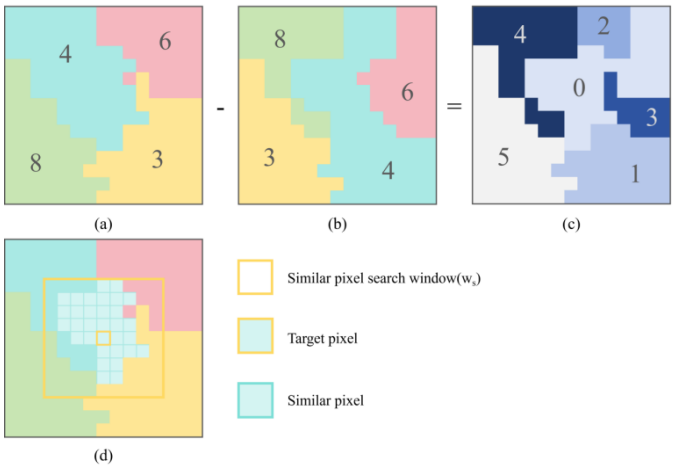


Fig. 2. Illustration of similar pixel selection under temporal consistency and spatial consistency. (a) Subsets of the clustered fine-resolution image at T_1. (b) Subsets of the clustered fine-resolution image at T_2. (c) Difference map between T_1 and T_2. (d) Schematic of similar pixel selection.

To preserve structural integrity during this correction, a spatial constraint based on object-level continuity is introduced. Pixels within the same segmented object exhibit similar temporal behavior and are therefore more likely to share similar residuals [33], [37]. Leveraging this property, OR prevents residual leakage across object boundaries while retaining sufficient flexibility to model within-object variation [38], [39]. The conceptual framework is illustrated in Fig. 2.

Ideally, similar pixels could be identified using true fine-resolution observations at T_2 to construct accurate temporal change trajectories. In practice, since fine-resolution data at T_2 are unavailable, HRSF employs the up-sampled coarse-resolution image as a surrogate. Although it lacks fine spatial details, it retains essential class information, allowing constrained residual estimation and redistribution within the proposed framework. The residual compensation within each segmented object is thus formulated as follows.

The spatial distance between the jth fine-resolution pixel and the central pixel of the ith coarse pixel is defined as

$$I_{distan}\left(x_{ij}, y_{ij}\right) = 1 + \sqrt{\left(x_{ij} - x_{i\frac{s^2}{2}}\right)^2 + \left(y_{ij} - y_{i\frac{s^2}{2}}\right)^2} \Big/ \frac{s}{2} \quad (4)$$

where $I_{distan}(x_{ij}, y_{ij})$ represent the indices used to measure spatial distance. $I_{distan}(x_{ij}, y_{ij})$ ranges between 1 and $1 + \sqrt{2}$. Here, s denotes the spatial resolution ratio between the fine and coarse-resolution pixels. The temporal-series difference between the jth fine-resolution pixel of the ith coarse-resolution pixel is defined as

$$valid\left(x_{ij}, y_{ij}\right) = \frac{1}{n_b}\sum_{b=1}^{n_b} C_{\mathrm{T}_2}\left(x_i, y_i, b\right) - C_{\mathrm{T}_1}\left(x_i, y_i, b\right) \quad (5)$$

$$I_{valid}\left(x_{ij}, y_{ij}\right) = \frac{1}{s^2}\sum_{j=1}^{s^2}\left(1 + \left|valid\left(x_{ij}, y_{ij}\right) - valid\left(x_{i\frac{s^2}{2}}, y_{i\frac{s^2}{2}}\right)\right|\right)^{-1} \quad (6)$$

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where $valid(x_{ij}, y_{ij})$ denotes the temporal-series difference of the fine-resolution pixel, and $I_{valid}(x_{ij}, y_{ij})$ represent the indices used to measure temporal dissimilarity, respectively. $I_{valid}(x_{ij}, y_{ij})$ ranges between 0 and 1. Here, n_b denotes the number of spectral bands. Based on these metrics, the similarity index is defined as

$$I_{sim}\left(x_{ij}, y_{ij}\right) = \frac{1 - I_{valid}\left(x_{ij}, y_{ij}\right)}{I_{distan}\left(x_{ij}, y_{ij}\right)} \tag{7}$$

where $I_{sim}(x_{ij}, y_{ij})$ denotes the similarity index, with values ranging between 0 and 1, and a larger value indicates higher similarity. Considering that fine-resolution images acquired at T_1 and T_2 generally exhibit similar spatial structures [40], the top $r\%$ of fine-resolution pixels, ranked by similarity indices within each segmented object, are selected as reference samples for residual estimation. Accordingly, the final fine-scale residual derived from OR is calculated as

$$\Delta F^{\mathrm{OR}}\left(x_{ij}, y_{ij}, b\right) = \sum_{r=1}^{r_s} \frac{I_{sim}}{\sum_{r=1}^{r_s} I_{sim}} \times R^{\mathrm{TR}}\left(x_{ij}, y_{ij}, b\right) \tag{8}$$

where $R^{\mathrm{TR}}(x_{ij}, y_{ij}, b)$ denotes the residual obtained by subtracting the downsampled $F^{\mathrm{TR}}_{T_2}(x_{ij}, y_{ij}, b)$ from the coarse-resolution image at T_2. $\Delta F^{\mathrm{OR}}(x_{ij}, y_{ij}, b)$ denotes the redistributed residual obtained by applying the OR to $R^{\mathrm{TR}}(x_{ij}, y_{ij}, b)$, and r_s represents the number of selected similar fine-resolution pixels within the segmented object. Accordingly, the prediction of OR is calculated as

$$F^{\mathrm{OR}}_{T_2}\left(x_{ij}, y_{ij}, b\right) = F^{\mathrm{TR}}_{T_2}\left(x_{ij}, y_{ij}, b\right) + \Delta F^{\mathrm{OR}}\left(x_{ij}, y_{ij}, b\right). \tag{9}$$

E. Context-Aware Residual Compensation

Although the OR stage suppresses cross-boundary contamination, an object-level support may still be too restrictive to recover local land-cover changes occurring within an object. Therefore, (2) applies the context residual compensation operator R_{G_C} to the updated OR prediction, thereby incorporating the neighborhood information of the target fine-resolution pixels into the reconstruction process. Similar to the strategy adopted in the OR stage, this step enhances reflectance estimation by exploiting similar pixels associated with the target pixel. The key distinction lies in that these similar pixels are no longer restricted to objects but are instead selected from a local window. To quantify the similarity between each neighboring pixel and the target pixel, a spectral distance index and a temporal difference index are introduced.

At the fine-resolution pixel scale, the reflectance change of the target pixel from T_1 to T_2 is defined as follows:

$$\Delta\left(x_{ij}, y_{ij}, b\right) = C_{T_2}\left(x_{ij}, y_{ij}, b\right) - F_{T_1}\left(x_{ij}, y_{ij}, b\right) \tag{10}$$

where $\Delta(x_{ij}, y_{ij}, b)$ denotes the reflectance change in band b, and $C_{T_2}(x_{ij}, y_{ij}, b)$ represents the up-sampled coarse-resolution image at T_2. It is important to note that both TR and OR were performed at the object level. Therefore, the temporal dissimilarity in these stages was computed at the coarse-pixel scale. In contrast, the CR is subsequently performed within a moving window centered on each target pixel, and the corresponding temporal dissimilarity is therefore computed at the fine pixel level. This strategy enables the context compensation module to distinguish fine-resolution pixels that are spatially adjacent but exhibit distinct temporal behaviors. Accordingly, the temporal dissimilarity between the kth neighboring pixel and the target pixel located at (x_{ij}, y_{ij}) is defined as

$$V_k = \left| \frac{1}{n_b} \sum_{b=1}^{n_b} \Delta\left(x_{ij}, y_{ij}, b\right) - \frac{1}{n_b} \sum_{b=1}^{n_b} \Delta\left(x_{ik}, y_{ik}, b\right) \right|. \tag{11}$$

The spectral distance between the kth neighboring pixel and the target pixel at (x_{ij}, y_{ij}) is defined as

$$S_k = \left| \frac{1}{n_b} \sum_{b=1}^{n_b} F_{T_1}\left(x_{ij}, y_{ij}, b\right) - \frac{1}{n_b} \sum_{b=1}^{n_b} F_{T_1}\left(x_{ik}, y_{ik}, b\right) \right|. \tag{12}$$

To jointly enforce spectral and temporal consistency, HRSF adopts a dual-ranking-based similar pixel selection strategy. For each target pixel, spectral similarity and temporal change consistency are independently measured within a local window, and candidate pixels are ranked accordingly, where higher ranks indicate greater similarity. A ranking-consistency constraint is then introduced to ensure reliability. A pixel is selected only if it ranks high in both domains and the ranking difference remains within a predefined threshold. This design reliably eliminates candidates with one-sided similarity, reducing mismatches and improving the robustness of similar-pixel selection. The constraint is formulated as follows:

$$Score_k = \begin{cases} \left(w_s^2 - r\left(V_k\right)\right) + \left(w_s^2 - r\left(S_k\right)\right), \\ \qquad\qquad |r\left(V_k\right) - r\left(S_k\right)| \le r_t \\ 0, \qquad\quad |r\left(V_k\right) - r\left(S_k\right)| > r_t \end{cases} \tag{13}$$

where $r(V_k)$ and $r(S_k)$ denote the ranking positions of the kth neighboring pixel in the change similarity ranking and spectral similarity ranking, respectively. r_t denotes the ranking-consistency threshold, which is related to the homogeneity of land-cover types within the w_s window and is typically set equal to the number of selected similar pixels k_s; the determination of the optimal value of r_t will be discussed in Section V. $Score_k$ represents the spectral-temporal consistency score of the kth neighboring pixel. Within a moving window of size $w_s \times w_s$ centered on the target pixel, the residuals of the top k_s neighboring pixels with the highest $Score_k$ values are selected and weighted according to their spatial distance. The spatial distance of the kth neighboring pixel is defined as

$$D_k = 1 + \sqrt{\left(x_{ij} - x_{ik}\right)^2 + \left(y_{ij} - y_{ik}\right)^2} \Big/ \frac{w_s}{2}. \tag{14}$$

Using the residual recomputed from the OR prediction, the context-aware compensation is then computed as

$$\Delta F^{\mathrm{CR}}\left(x_{ij}, y_{ij}, b\right) = \sum_{k=1}^{k_s} \frac{1/D_k}{\sum_{k=1}^{k_s} 1/D_k} \times R^{\mathrm{OR}}\left(x_{ij}, y_{ij}, b\right) \tag{15}$$

and the final prediction result is given by

$$F^{\mathrm{CR}}_{T_2}\left(x_{ij}, y_{ij}, b\right) = F^{\mathrm{OR}}_{T_2}\left(x_{ij}, y_{ij}, b\right) + \Delta F^{\mathrm{CR}}\left(x_{ij}, y_{ij}, b\right). \tag{16}$$

F. Confidence-Guided Adaptive Fusion

Sections II-C–II-E produce a hierarchical residual estimate by progressively refining the homogeneity support from classes to objects and local pixel neighborhoods. However, the reliability of this estimate varies with temporal-change magnitude and local structural complexity. To better balance

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spectral fidelity and spatial accuracy under heterogeneous conditions, a *CSII* is proposed. This index jointly characterizes the influence of reflectance change magnitude and spatial information content on the final prediction, enabling adaptive weighting across different surface conditions. Specifically, the *CSII* integrates two complementary components. The first component quantifies the magnitude of reflectance change, capturing the intensity of temporal variation and highlighting regions with significant dynamics. The second component measures the spatial information content of the target pixel, reflecting local structural complexity and the availability of fine-scale details. By combining these two factors, the proposed index adaptively adjusts the contribution of residual redistribution, promoting spectral consistency in stable regions while enhancing spatial detail preservation in heterogeneous areas.

The spatial information of the target pixels is quantified as [41]

$$SIR\left(x_{ij},y_{ij},b\right)=\frac{\left|RE\left(x_{ij},y_{ij},b\right)-E\left[RE\left(x_{ij},y_{ij},b\right)\right]\right|}{4\times Stddev\left[RE\left(x_{ij},y_{ij},b\right)\right]} \quad (17)$$

where $RE(x_{ij},y_{ij},b)$ denotes the Roberts edge response in band b, and $Stddev[]$ represents the standard deviation. Values exceeding 1 are truncated to 1, where larger values indicate that the pixel is surrounded by richer spatial structural information. Based on this index, the *CSII* for the target pixel is defined as follows:

$$CSII\left(x_{ij},y_{ij},b\right)=\frac{\left[\Delta\left(x_{ij},y_{ij},b\right)-min\left[\Delta\left(x_{ij},y_{ij},b\right)\right]\right]}{\left[max\left[\Delta\left(x_{ij},y_{ij},b\right)\right]-\Delta\left(x_{ij},y_{ij},b\right)\right]}\times\left[1-SIR\left(x_{ij},y_{ij},b\right)\right]. \quad (18)$$

Accordingly, the final result is formulated as

$$F_{T_2}^{CF}\left(x_{ij},y_{ij},b\right)=CSII\left(x_{ij},y_{ij},b\right)\times F_{T_2}^{BIC}\left(x_{ij},y_{ij},b\right)+\left[1-CSII\left(x_{ij},y_{ij},b\right)\right]\times F_{T_2}^{CR}\left(x_{ij},y_{ij},b\right). \quad (19)$$

To facilitate reproducibility and improve clarity, the overall workflow of the proposed HRSF framework is summarized in Algorithm 1.

Algorithm 1 HRSF Framework

Input: Fine-resolution image F_{T_1}, Coarse-resolution images C_{T_1} and C_{T_2}

Output: Predicted fine-resolution image $F_{T_2}^{CF}$

// Step 1: Preprocessing (cross-sensor correction) For each object:
  Perform unsupervised classification on F_{T_1}
  Fit class-wise quadratic regression models
  For each pixel:
    Correct C_{T_1} and C_{T_2} using the corresponding class models
  End for
// Step 2: Temporal Residual Estimation (TR)
  Estimate initial fine-resolution prediction $F_{T_2}^{TF}$ using temporal modeling
// Step 3: Object-constrained Residual Compensation (OR)
  For each pixel:
    Compute residuals R at the coarse-resolution pixel level
    Identify similar pixels using similarity index
    Redistribute residuals based on similarity index
  End for
End for
// Step 4: Context-aware Residual Compensation (CR)
  For each pixel:
    Identify similar pixels using dual-ranking strategy Redistribute residuals based on spatial distance
// Step 5: Confidence-Guided Adaptive Fusion (CF)
  Compute adaptive weights (*CSII*)
  Adaptively fuse CR result and Bicubic-interpolated coarse-resolution image at T2 using *CSII*
End for

III. Experiments and Results

A. Study Sites and Experimental Setup

Six-band image pairs (blue, green, red, NIR, SWIR1, and SWIR2) were used for the spatiotemporal fusion experiments. To evaluate the robustness and general applicability of the proposed method under diverse surface conditions, all experiments were performed on a workstation with an Intel Core i7-13650HX CPU (2.60 GHz). For images of size 1026 × 1026 pixels, the proposed HRSF method requires approximately 3.5 min per image on average. The reported runtime refers only to the fusion process and does not include data preprocessing. As illustrated in Fig. 3, four study areas with distinct characteristics were selected, representing highly heterogeneous landscapes with strong temporal variations (CIA), relatively homogeneous regions with simple structures (LGC) [42], structured heterogeneous agricultural areas with regular spatial patterns (IC), and complex coastal-desert landscapes with fine-scale spatial variations (Mas). To further examine the impact of cross-sensor discrepancies, three types of coarse-resolution data were employed: original, preprocessed, and simulated imagery. The preprocessed data were used to reduce systematic sensor differences, while the simulated data, generated by aggregating fine-resolution imagery, eliminated input uncertainties and satisfied the assumptions of spatiotemporal fusion [22].

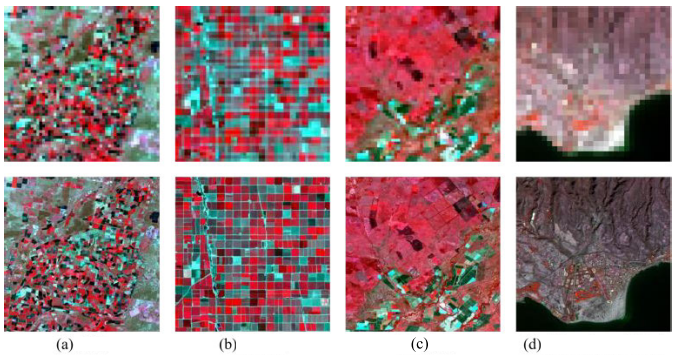


Fig. 3. Study area of this study. (a) CIA site. (b) IC site. (c) LGC site. (d) Maspaloma (Mas). (Top row) Simulated MODIS image at T_1. (Bottom row) Corresponding Landsat image at T_1.

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TABLE I
PARAMETER SETTINGS FOR COMPARISON METHODS AND HRSF. N/A INDICATES THAT THE PARAMETER IS NOT APPLICABLE TO THE METHOD

Method	r_t		Class	w	w_s	k_s	r%
	CIA/LGC /Mas	IC site					
FSDAF	N/A	N/A	5	N/A	31	20	N/A
Fit-FC	N/A	N/A	N/A	3	31	20	N/A
FSDAF2.0	N/A	N/A	5	N/A	31	20	N/A
RASDF	N/A	N/A	3-7	N/A	N/A	N/A	N/A
VSDF	N/A	N/A	5	N/A	N/A	20	N/A
First	N/A	N/A	N/A	N/A	N/A	N/A	N/A
OL-HSTFM	N/A	N/A	N/A	N/A	N/A	N/A	N/A
OBSUM	N/A	N/A	5	15	31	20	5
HRSF	20	70	5	15	31	20	5

To validate the effectiveness of the proposed method, eight representative spatiotemporal fusion methods that also use a single fine-coarse image pair as input were selected for comparison, including OBSUM, OL-HSTFM, First, VSDF, RASDF, FSDAF2.0, Fit-FC, and FSDAF. These methods have been widely used in spatiotemporal fusion studies and require only one complete fine-coarse-resolution image pair at the reference date and one coarse-resolution image at the prediction date. The corresponding parameter settings for comparison methods and HRSF are summarized in Table I. Parameters were determined according to the recommendations in the original studies, while common parameters shared by different methods were assigned identical values for consistency and fair comparison. For methods requiring segmentation information, segmentation maps generated by the Segment Anything Model were used [43]. By default, classification maps were generated using K-means clustering [44].

For the quantitative evaluation of fusion performance, four metrics were employed: root-mean-square error (RMSE), mean absolute error (MAE), correlation coefficient (r), and local binary pattern (LBP), which evaluate spectral accuracy and local texture consistency, respectively. RMSE and MAE measure the deviation between the predicted and reference reflectance values, while r quantifies their linear correlation and LBP characterizes the similarity of local texture patterns. For consistent comparison among methods, all metrics were computed as mean values across the six spectral bands, and the absolute values of MAE and LBP are reported. In the tables, upward and downward arrows (↑/↓) indicate that higher or lower values correspond to superior performance, respectively.

B. Comparison of Preprocessing Strategies

Most spatiotemporal fusion methods employ regression-based preprocessing to reduce discrepancies between multisource images. In this study, four commonly used strategies were evaluated: global linear regression, moving-window linear regression, class-based linear regression, and class-based quadratic polynomial regression.

TABLE II
ACCURACY METRICS FOR DIFFERENT PREPROCESSING METHODS

Site and date	Metric	Linear	Window Linear	Class Linear	Class Poly
CIA Feb. 12,2002	RMSE↓	0.0830	0.3032	0.0870	**0.0611**
	MAE↓	0.0625	0.1259	0.0673	**0.0482**
	r↑	0.5156	0.3656	0.4163	**0.6796**
	LBP↓	**0.0000**	0.0006	**0.0000**	**0.0000**
LGC Aug. 22,2004	RMSE↓	0.0445	0.0868	0.0474	**0.0441**
	MAE↓	**0.0325**	0.0696	0.0357	0.0339
	r↑	0.7870	0.7162	0.7745	**0.8343**
	LBP↓	**0.0000**	**0.0000**	**0.0000**	**0.0000**
IC site June 25,2023	RMSE↓	0.0724	0.1778	0.0684	**0.0581**
	MAE↓	0.0560	0.1138	0.0517	**0.0447**
	r↑	0.8181	0.4783	0.8112	**0.8548**
	LBP↓	**0.0000**	**0.0000**	**0.0000**	**0.0000**

The comparison results are presented in Table II. The bold values indicate the highest accuracy, while the underlined values denote the second-highest accuracy in each metric. Since the simulated MODIS imagery is generated from Landsat data through spatial aggregation, systematic discrepancies between the two sensors are not expected. Accordingly, the same evaluation framework described above was adopted to assess the spectral consistency and local texture similarity between the corrected coarse-resolution image at T_2 and the simulated MODIS image. As shown in Table II, class-based quadratic polynomial regression consistently achieved the best or near-best performance for most evaluation metrics across all three study areas. In the highly heterogeneous CIA and IC sites, the RMSE values are lower than those of the second-best method by 0.0143 and 0.0103, respectively, whereas the improvement in the relatively homogeneous LGC site is only 0.0003.

C. Contributions of Individual Modules in HRSF

To minimize the influence of cross-sensor discrepancies and other systematic factors on the assessment of the proposed method's intrinsic capability, both real satellite imagery and simulated image pairs were employed. The four components of the proposed framework are denoted as TR, OR, context-aware residual compensation (CR), and confidence-guided adaptive fusion (CF). These modules operate sequentially, enabling progressive refinement of residual estimation and redistribution. Accordingly, ablation experiments were designed to evaluate their cumulative contributions rather than individual performance. To assess robustness under different temporal conditions, simulated image pairs with varying time intervals were selected across the CIA, IC, and LGC sites. The resulting RMSE values (CIA: 0.1779, IC: 0.2142, and LGC: 0.1165) indicate stronger temporal variation in CIA and IC compared with LGC, providing diverse scenarios for evaluating module effectiveness.

Fig. 4 shows the intermediate fusion results produced by the proposed method in comparison with the corresponding reference fine-resolution images. As we progress through the four stages, the fused images progressively approach the reference images. To illustrate the contribution of each module,

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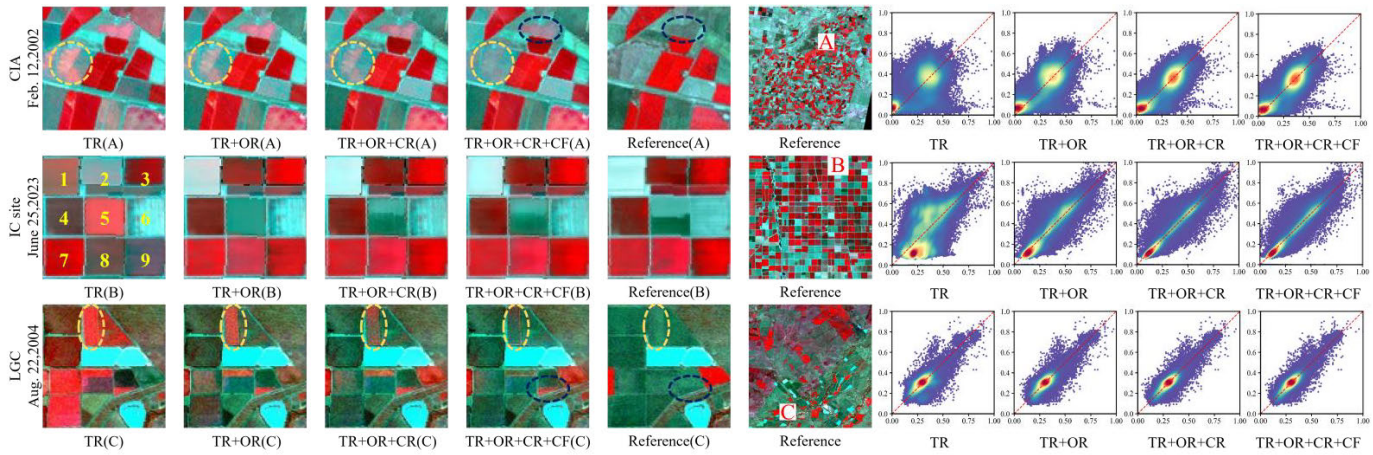


Fig. 4. Progressive refinement of HRSF in three sites, from four distinct steps. The four left columns show scatter plots of the four-step results versus the red band of the reference image. Yellow- and black-circled areas highlight regions where the results differ from the reference. (a) CIA site. (b) IC site. (c) LGC site.

TABLE III
ACCURACY METRICS FOR THE FOUR INTEGRATION STEPS OF HRSF

Site and date	Metric	TR	TR+OR	TR+CR	TR+OR +CR	TR+OR +CR+CF
CIA Feb. 12,2002	RMSE↓	0.1226	0.0950	0.0826	0.0773	**0.0723**
	MAE↓	0.0863	0.0657	0.0580	0.0532	**0.0505**
	r↑	0.6342	0.7926	0.8436	0.8662	**0.8761**
	LBP↓	**0.0025**	0.0033	0.0049	0.0056	0.0046
LGC Aug. 22,2004	RMSE↓	0.0758	0.0526	0.0476	0.0443	**0.0426**
	MAE↓	0.0510	0.0366	0.0328	0.0303	**0.0294**
	r↑	0.7554	0.8835	0.9042	0.9180	**0.9219**
	LBP↓	**0.0004**	0.0005	0.0042	0.0033	0.0031
IC site June 25,2023	RMSE↓	0.1463	0.0741	0.0778	0.0602	**0.0585**
	MAE↓	0.1149	0.0515	0.0542	**0.0377**	0.0382
	r↑	0.5626	0.9039	0.8937	0.9383	**0.9411**
	LBP↓	0.0066	0.0066	0.0128	0.0041	**0.0033**

the IC site is selected as an example. In the TR, several regions (blocks 3, 4, 6, and 7) already resemble the reference image, while others exhibit spectral unmixing errors due to land-cover changes. After OR, residuals are redistributed within object boundaries, effectively reducing spectral distortions. However, since TR and OR operate at the object level, fine-scale details remain insufficient. The CR stage further incorporates neighborhood information to restore within-object variations, particularly in changed regions (e.g., block 5), producing results closer to the reference image. In the CF, constraints based on spatial information richness and change magnitude, further improving prediction accuracy. This progressive improvement is confirmed by the RED-band scatter plots in Fig. 4, which gradually converge toward the 1:1 line. Overall, all modules contribute cumulatively to performance gains. However, some errors remain (blue circles in CIA and LGC), mainly in small-scale change areas that cannot be reliably captured from coarse-resolution observations at T_2.

Table III presents the evaluation results for different module combinations, illustrating the progressive improvement of the proposed framework. Across all study areas, RMSE decreases significantly when transitioning from TR to TR + OR, indicating enhanced spectral consistency. At the IC site, LBP decreases slightly after introducing OR, suggesting that OR suppresses spatial blurring. In contrast, LBP increases slightly at the CIA and LGC sites by 0.0011 and 0.0001, respectively, mainly due to spectral correction altering inherited texture patterns. Nevertheless, RMSE is reduced by 0.0276 and 0.0232 at the CIA and LGC sites, confirming improved spectral accuracy. When OR is removed (TR + CR), residual propagation relies solely on neighborhood information, leading to oversmoothing and blurred boundaries. As a result, LBP increases markedly. When CR is introduced after OR (TR + OR + CR), overall consistency is further improved, as neighborhood-based redistribution enhances the modeling of within-object variations. However, the moving-window strategy may still oversmooth local textures, resulting in an increase in LBP. Finally, the CF module further incorporates spectral information from T_2 for adaptive refinement, further improving both spectral accuracy and spatial consistency. These results demonstrate that each module contributes incrementally, and their integration yields the best overall performance.

D. Comparison With Other Methods

To facilitate a direct comparison between HRSF and the comparative methods, the same image pairs as described in Section III-C are used. Visual inspection of the full scenes indicates that all methods can reconstruct the T_2 image to varying degrees.

To further illustrate fusion details, Fig. 5 shows enlarged comparisons of the subareas highlighted by white boxes in the first column. In Fig. 5(a), most methods fail to accurately capture land-cover changes within the yellow-circled area. In contrast, the proposed method maintains sharp land-cover boundaries by leveraging object-level unmixing and residual redistribution. Moreover, through the adaptive weighting module, which incorporates spectral information at T_2, the spectral distortions observed in other methods are successfully mitigated. In Fig. 5(b), Fit-FC and FSDAF exhibit noticeable smoothing of land-cover edges, while Fit-FC produces inaccurate predictions for a road segment in the lower right corner. Although OBSUM, OL-HSTFM, First, VSDF, RASDF, and FSDAF2.0 partially preserve land-cover boundaries, they fail to accurately capture land-cover changes within

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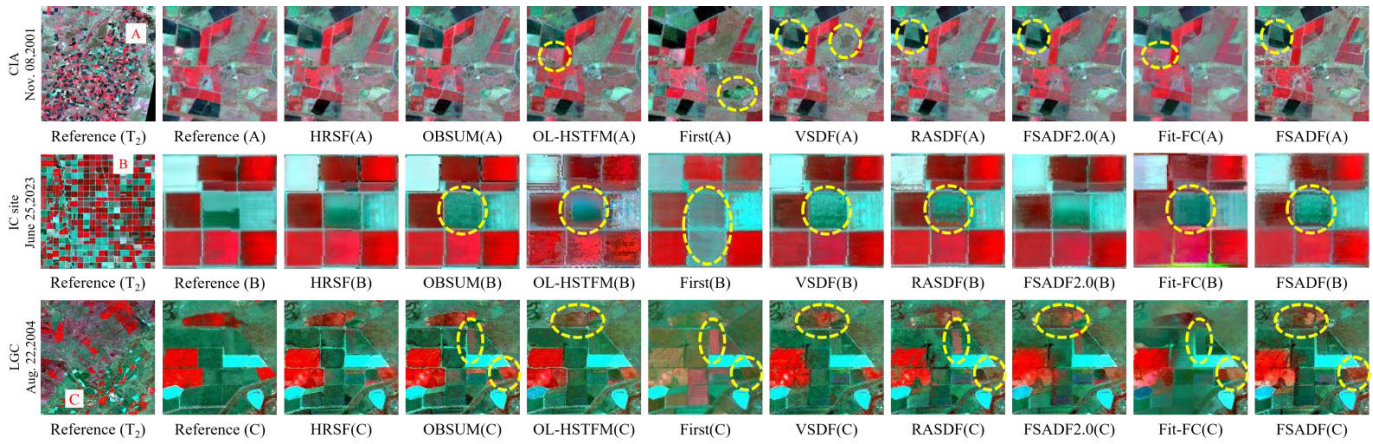


Fig. 5. Predictions and reference imagery from different spatiotemporal fusion methods for (a) CIA, (b) IC, and (c) LGC sites. Yellow-circled areas highlight regions where the results differ from the reference.

TABLE IV
ACCURACY METRICS FOR DIFFERENT SPATIOTEMPORAL FUSION METHODS

Site and date	Metric	FSDAF	Fit-FC	FSDAF2.0	RASDF	VSDF	First	OL-HSTFM	OBSUM	HRSF
CIA Nov. 08,2001	RMSE↓	0.0677	0.0756	0.0624	0.0674	**0.0607**	0.1036	0.0765	0.0651	0.0616
	MAE↓	0.0467	0.0511	0.0426	0.0443	**0.0407**	0.0692	0.0501	0.0430	0.0420
	r↑	0.8614	0.8263	0.8834	0.8629	**0.8870**	0.6402	0.8132	0.8724	0.8834
	LBP↓	0.0190	0.0131	0.0110	0.0080	0.0121	0.0264	0.0076	0.0056	**0.0036**
IC June 25,2023	RMSE↓	0.0805	0.1004	0.0699	0.0710	0.0642	0.1317	0.1052	0.0626	**0.0585**
	MAE↓	0.0577	0.0690	0.0499	0.0500	0.0440	0.1053	0.0812	0.0387	**0.0382**
	r↑	0.8824	0.8078	0.9165	0.9124	0.9285	0.6477	0.8164	0.9342	**0.9411**
	LBP↓	0.0370	0.0271	0.0100	0.0252	0.0239	0.0311	0.0182	0.0130	**0.0033**
LGC Aug. 22,2004	RMSE↓	0.0589	0.0511	0.0511	0.0526	0.0427	0.0776	0.0469	0.0446	**0.0426**
	MAE↓	0.0397	0.0361	0.0365	0.0357	0.0297	0.0583	0.0331	0.0303	**0.0294**
	r↑	0.8625	0.8929	0.8915	0.8816	0.9216	0.7340	0.9051	0.9174	**0.9219**
	LBP↓	0.0141	0.0042	0.0109	0.0115	0.0117	0.0347	0.0045	0.0039	**0.0031**

the yellow-circled area. Moreover, OL-HSTFM and RASDF exhibit pronounced salt-and-pepper spectral noise, indicating limited stability in sites undergoing change. By contrast, the proposed method integrates neighborhood information with temporal-difference awareness. By jointly enforcing explicit and implicit constraints, it prevents excessive smoothing along land-cover boundaries and better preserves spatial structures under change scenarios. In Fig. 5(c), Fit-FC fails to accurately capture land-cover changes within the yellow-circled area, while continuing to excessively smooth edges. OL-HSTFM, First, VSDF, RASDF, and FSDAF2.0 also fail to accurately capture the changed features within the yellow-circled area. By contrast, the proposed method and OBSUM both reconstruct the target features reasonably well, with the proposed method exhibiting superior spatial localization and morphological consistency. Overall, the results indicate that the proposed method constrains residual propagation by jointly modeling object-level unmixing and spatiotemporally constrained residual redistribution, imposing both explicit and implicit constraints and preventing excessive smoothing along land-cover boundaries. By incorporating neighborhood information within the objects, the method improves robustness against land-cover changes, while the integration of spectral information at T_2 suppresses spectral distortion, achieving a balanced tradeoff between spectral consistency and spatial structure preservation.

For the selected fusion tasks, HRSF achieves the best overall performance at the IC and LGC sites while remaining competitive at the CIA site. Table IV presents a quantitative comparison of nine spatiotemporal fusion methods across the three study areas. At the heterogeneous CIA site with irregular spatial structures, VSDF obtains the best RMSE, MAE, and r, whereas HRSF achieves the lowest LBP of 0.0036, outperforming OBSUM by 0.0020 and demonstrating superior structural preservation. At the IC site, characterized by strong heterogeneity and relatively regular parcel geometry, HRSF ranks first in all four metrics. It reduces RMSE by 0.0041 and increases r by 0.0069 compared with OBSUM, while its LBP is 0.0067 lower than that of FSDAF2.0. At the homogeneous LGC site, differences among the leading methods are relatively small, but HRSF still achieves the best results across all metrics. These results indicate that the advantages of HRSF are most pronounced in heterogeneous regions with regular spatial structures.

To further assess the spatial and spectral performance of different methods, Fig. 6 presents the classified maps and reference classified maps derived from local regions of the predicted images from the three study areas. The reference

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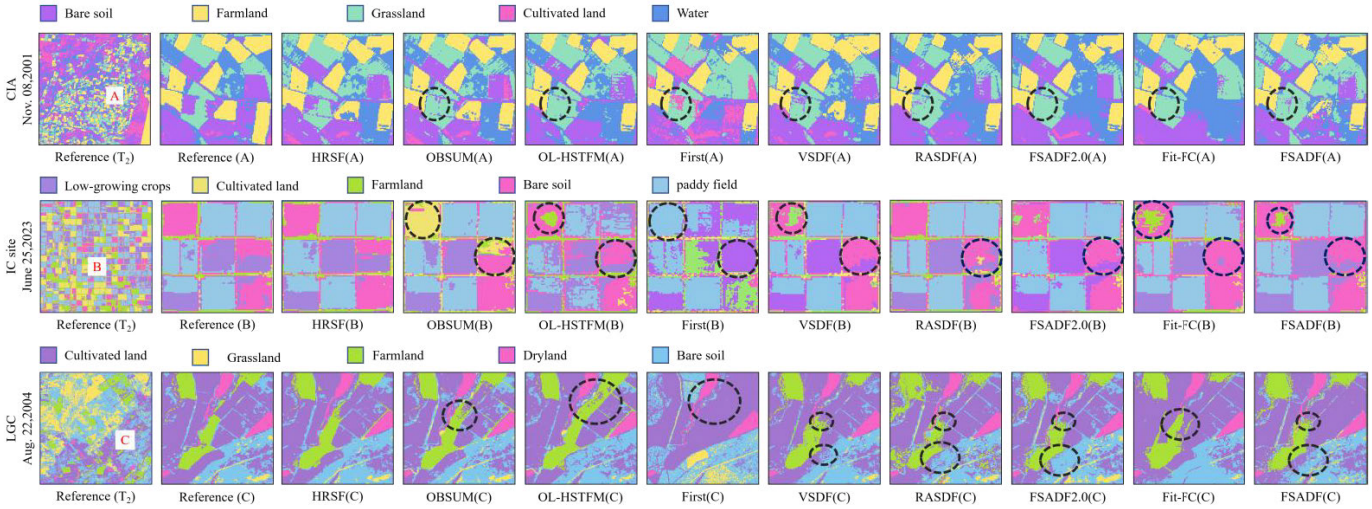


Fig. 6. Classification results from fusion methods and reference imagery for (a) CIA, (b) IC, and (c) LGC sites. Black-circled areas highlight regions where the classification results differ from the reference.

classification image is a classification map of the corresponding Landsat imagery generated using a support vector machine (SVM), and it is utilized for both quantitative and qualitative assessment. This map does not constitute an official land cover product but rather serves as a benchmark for evaluating spatial clustering consistency and the efficacy of spectral reconstruction.

In the IC site, the black-circled area in the enlarged subarea corresponds to cropland. Most methods correctly recover the land-cover category at the class level. However, compared with the reference classification map, the results of OL-HSTFM, VSDF, RASDF, FSDAF2.0, Fit-FC, and FSDAF exhibit noticeable spatial structural deviations. OBSUM and First produce a spatial pattern closer to the reference map but misclassify the land-cover type within the circled area. Similar patterns are observed in the CIA and LGC study areas. Only the proposed HRSF method accurately reconstructs the land-cover categories and spatial structures within the circled areas, achieving high consistency with the reference classification map.

E. Temporal Robustness of HRSF in Long Time Series

To evaluate the robustness of HRSF under long-term time-series conditions, its performance was assessed using multisensor input data from the Mas and LGC datasets. A total of 14 image pairs were selected from each dataset for the experiments. The Mas dataset consists of Sentinel-2 and Sentinel-3 image pairs, with a spatial resolution ratio of 30 between the coarse- and fine-resolution images, whereas the LGC dataset consists of Landsat and MODIS image pairs with a resolution ratio of 19. These experiments were designed to evaluate the adaptability of HRSF under different sensor combinations and long-term temporal observation scenarios.

Fig. 7 illustrates the temporal variation of the four evaluation metrics across the Mas and LGC sites. RMSE, MAE, and LBP exhibit consistent temporal patterns, while r shows an opposite trend, reflecting the complementary relationship between error-based and correlation-based metrics. The peaks of all metrics coincide with those of the reference RMSE between T_1 and T_2, indicating that prediction difficulty is primarily driven by the magnitude of land-cover change rather than the temporal interval alone. Despite similar temporal trends among different methods, HRSF generally achieves the lowest RMSE and MAE and the highest correlation coefficient r throughout the time series, indicating its superior ability to preserve spectral-temporal variations. Its advantages are particularly evident at error peaks, where it effectively suppresses error accumulation during long-term prediction. However, HRSF does not consistently attain the lowest LBP in the Mas dataset. This is mainly because Section II-F introduces the interpolated coarse-resolution image at T_2. Given the large coarse-to-fine spatial resolution ratio of 30, uncertainties in the coarse-resolution information may affect the reconstruction of fine-scale spatial structures. Meanwhile, object-based fusion methods such as OBSUM and OL-HSTFM better preserve local spatial structures by constraining residual propagation within segmented objects, resulting in lower LBP values. However, this stronger structural preservation is accompanied by reduced spectral similarity, as reflected by their inferior RMSE, MAE, and correlation coefficient compared with HRSF. In contrast, for the LGC dataset, with a smaller resolution ratio of 19, HRSF achieves the best performance across all metrics with smaller temporal fluctuations, demonstrating strong stability under long-term time-series conditions.

These temporal patterns are further supported by the average results presented in Table V. In the LGC dataset, HRSF achieves the best performance across all four metrics, reducing RMSE and LBP by 0.0007 and 0.0015, respectively, compared with the second-best methods. In the Mas dataset, HRSF obtains the lowest RMSE, the joint-lowest MAE, and the highest r, although its LBP remains higher than those of OL-HSTFM and OBSUM. Overall, these results demonstrate that HRSF maintains high spectral reconstruction accuracy and temporal consistency across datasets.

IV. Application of HRSF in Flood Monitoring and Mapping

Section III demonstrated that HRSF can accurately reconstruct pronounced temporal variations and effectively recover

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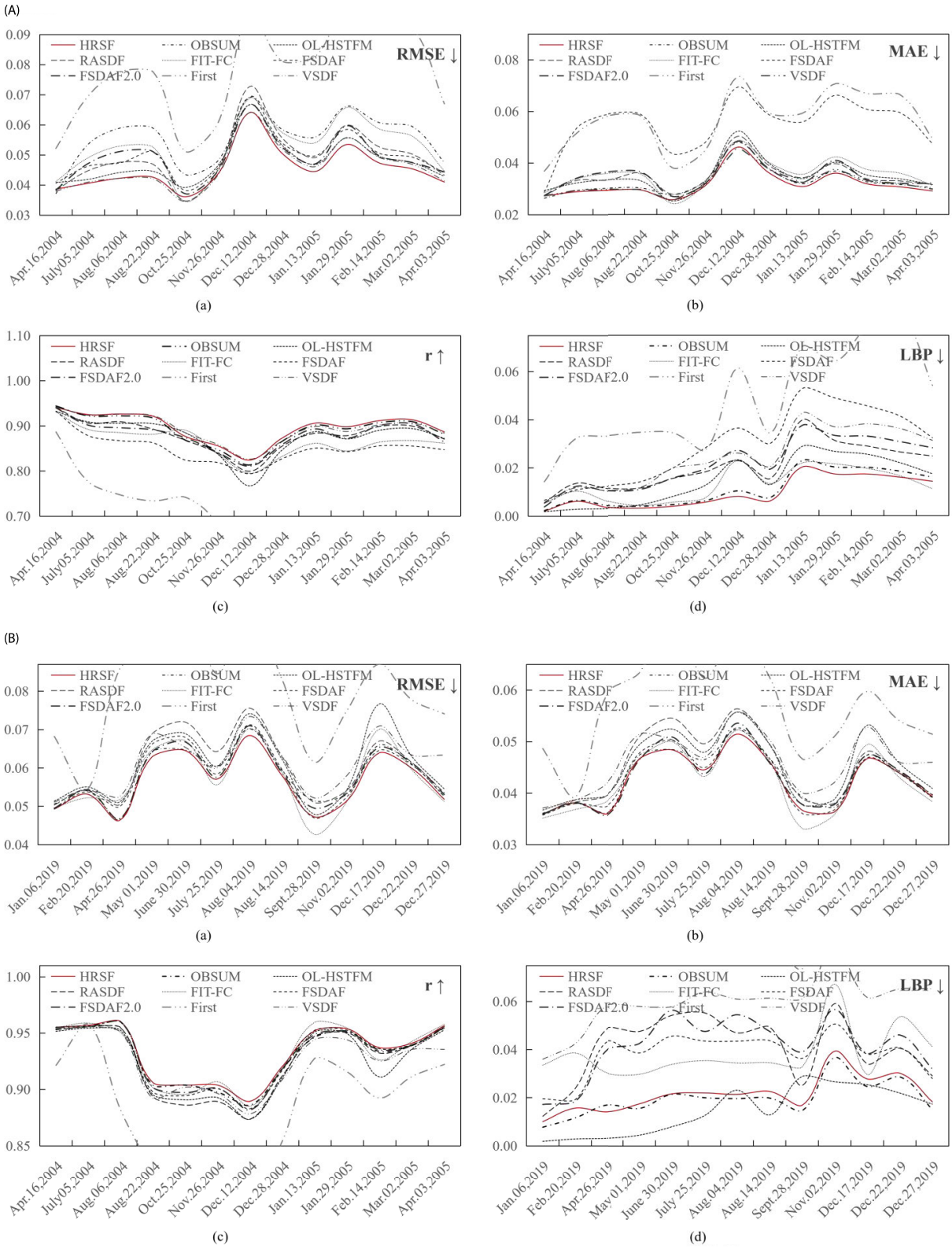


Fig. 7. Time-series trend curves of four accuracy metrics across all time frames except T_1 in the LGC, Mas site for different fusion methods. (A) LGC. (B) Mas. (a) RMSE. (b) MAE. (c) *r*. (d) LBP.

land-cover changes. To further assess the practical utility of HRSF, a severe flood event in Normanton, northwestern Queensland, Australia, was selected as a case study to monitor flood dynamics. Due to the extensive inundation, a subarea measuring 1275 × 1275 pixels was extracted for analysis, corresponding to an area of approximately 1463 km^2. The

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TABLE V
AVERAGE ACCURACY METRIC OF ALL PREDICTION RESULTS ACROSS TWO SITES USING DIFFERENT SPATIOTEMPORAL FUSION METHODS

Site	Metric	FSDAF	Fit-FC	FSDAF2.0	RASDF	VSDF	First	OL-HSTFM	OBSUM	HRSF
LGC	RMSE↓	0.0555	0.0524	0.0490	0.0494	0.0459	0.0772	0.0495	0.0474	**0.0452**
	MAE↓	0.0381	0.0355	0.0346	0.0348	0.0324	0.0564	0.0348	0.0328	**0.0316**
	r↑	0.8506	0.8651	0.8808	0.8804	0.8947	0.6680	0.8773	0.8920	**0.8977**
	LBP↓	0.0292	0.0128	0.0217	0.0204	0.0240	0.0480	0.0146	0.0110	**0.0095**
MAS	RMSE↓	0.0610	0.0603	0.0608	0.0635	0.0643	0.0838	0.0636	0.0611	**0.0595**
	MAE↓	0.0378	**0.0373**	0.0379	0.0399	0.0412	0.0524	0.0401	0.0380	**0.0373**
	r↑	0.9305	0.9318	0.9309	0.9264	0.9248	0.8766	0.9248	0.9313	**0.9337**
	LBP↓	0.0376	0.0381	0.0413	0.0406	0.0598	0.0912	**0.0110**	0.0195	0.0214

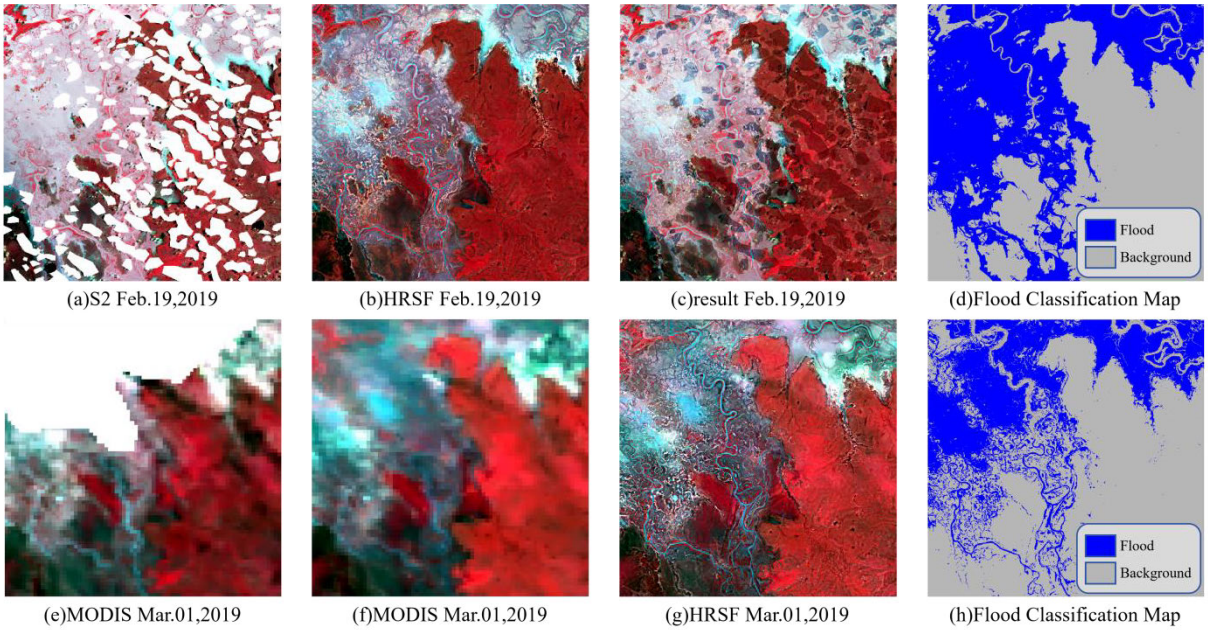


Fig. 8. Example flood monitoring results for February 19, 2019 and March 1, 2019. (a) Cloud-masked Sentinel-2 image on February 19, 2019. (b) HRSF-predicted image for February 19, 2019. (c) Fusion result combining HRSF prediction with cloud-masked Sentinel-2 image. (d) Flood extent derived from (c). (e) Cloud-masked MODIS image on March 1, 2019. (f) MODIS image on February 28 used for gap filling. (g) HRSF-predicted image for March 1, 2019. (h) Flood extent derived from (g).

observation period spanned 26 days, from February 9, 2019 to March 6, 2019. During this period, 16 usable MOD09GA images were acquired, along with a complete Sentinel-2 observation on March 6. In addition, three Harmonized Sentinel-2 MSI Level-2A images and Landsat 8/9 Collection 2 Level-2A images were employed as auxiliary datasets. However, these auxiliary images were partially affected by clouds and contained incomplete orbital coverage. Due to persistent rainfall in the days preceding the flood event, the MODIS and Sentinel-2 images acquired on December 26, 2018 were chosen as the T_1 reference images.

All images were reprojected to the geographic coordinate system (GCS) with the WGS 1984 datum. All bands of the Sentinel-2 and Landsat images were resampled to a spatial resolution of 30 m, whereas MODIS images were resampled to 510 m. The flood extent was delineated using a SVM-based supervised classification, with permanent water bodies masked to prevent interference. Fig. 8 illustrates example results for February 19 and March 1. Fig. 8(a) shows the cloud-masked Sentinel-2 image, whereas Fig. 8(c) presents the fusion result obtained by integrating the HRSF prediction with this image. Fig. 8(e) shows the cloud-masked MODIS image. After filling missing values using the MODIS observation acquired on February 28, satisfactory prediction results were obtained, as shown in Fig. 8(g). The corresponding flood monitoring results are presented in Fig. 8(d) and (h).

Thirteen temporally continuous prediction images were generated using the spatiotemporal fusion framework, substantially enhancing temporal resolution and providing continuous observations for dynamic flood monitoring. To characterize flood evolution, flood inundation was delineated for each fused image, enabling continuous monitoring of inundation dynamics. The flood area was subsequently calculated for each observation time. Fig. 9 shows that cubic spline interpolation was employed to interpolate missing observation dates, indicated by the orange-yellow points, thereby maintaining the temporal continuity of the time series.

The flood area shows a stepwise decline over time, indicating that the recession process progressed through multiple stages. The fit exponential trend line exhibits a consistent downward trajectory, indicating a monotonic attenuation of flood extent during the recession process. Temporally, the flood

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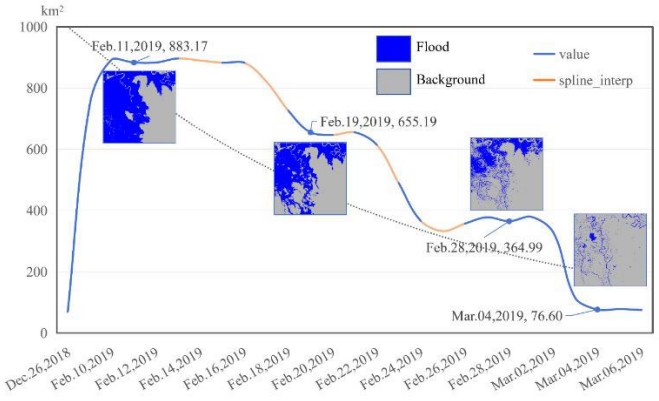


Fig. 9. Temporal evolution of flood extent. The chart shows flood-affected areas extracted on February 11, February 19, February 28, and March 4, 2019. Blue indicates inundated areas.

extent stabilized around March 4, 2019, indicating that the flood had progressed into the late recession stage. The water extent extracted from the full Sentinel-2 observation acquired on March 6, 2019 closely approximates the baseline water extent derived from December 26, 2018, indicating that the inundated area had largely returned to normal hydrological conditions. Based on this temporal analysis, the flood duration within the selected subarea is estimated at approximately 24 days, starting from the first observation date.

To quantitatively assess the applicability of the predicted images for flood detection, the emergency flood mapping products for February 9 and 13, 2019 provided by the Copernicus Emergency Management Service (EMS) were adopted as reference data. Fig. 10 illustrates the spatial comparison of flood extraction results using the predicted images produced by different spatiotemporal fusion methods against the reference image. In this experiment, identical training samples were applied to all classifications, and the SVM parameters were maintained unchanged during flood extraction, thereby minimizing misclassification effects arising from the classification procedure. Overall, the predicted results generally reconstruct the spatial distribution pattern of the flood with reasonable accuracy. The red-circled area highlights several misclassification cases observed in the extraction results produced by different methods. In contrast, the inundation boundaries and local details derived from the HRSF-predicted images remain largely consistent with the ground truth, indicating that the fused images reliably represent spatial structures. Notably, local inconsistencies are evident in the region highlighted by the yellow box. Further analysis indicates that the area was affected by cloud cover on February 9, resulting in incomplete observations in the reference product at that location, and causing deviations from the predicted results. Hence, this discrepancy primarily arises from the observational limitations of the reference data rather than errors in model predictions, which is reasonable in the context of remote sensing applications.

A quantitative comparison of flood extraction results derived from different predicted images is presented in Table VI. Compared with the ground-truth map, the flood detection results obtained from the HRSF-predicted images achieve the highest OA and Kappa, at 90.53% and 0.8093, respectively.

TABLE VI
ACCURACY METRICS FOR FLOOD EXTENT MAPS DERIVED FROM FUSION IMAGERY

Method	OA	Kappa	F1	Recall
HRSF	**90.53%**	**0.8093**	**0.9122**	0.9114
OBSUM	83.75%	0.6707	0.8549	0.8873
OL-HSTFM	85.20%	0.6979	0.8730	0.9425
First	77.81%	0.5505	0.8018	0.8314
VSDF	84.37%	0.6813	0.8653	0.9295
RASDF	84.92%	0.6919	0.8713	**0.9454**
FSDAF2.0	84.35%	0.6792	0.8642	0.9272
Fit-FC	85.01%	0.6946	0.8703	0.9319
FSDAF	83.87%	0.6714	0.8601	0.9191

The $F1$-score and recall reach 0.9122 and 0.9114, indicating high accuracy in identifying the flood class. The combination of spatial comparisons and quantitative metrics demonstrates that HRSF attains superior accuracy and robustness in flood monitoring applications, highlighting its potential for practical disaster monitoring.

V. DISCUSSION

A. Effectiveness of Preprocessing Methods

To assess the effectiveness of the class-based quadratic polynomial fitting preprocessing strategy, additional experiments were performed in the CIA and LGC study areas. All temporal scenes within each dataset were employed, and time-series trend analyses were performed on the fusion results obtained with and without preprocessing. Overall, during the first two months of the time series, the fusion results obtained with preprocessing demonstrate significantly higher accuracy than those obtained without preprocessing. In the later stages, however, the results obtained without preprocessing exhibit slightly higher accuracy. This pattern is closely related to the inherent preprocessing characteristics of the datasets. Prior to release, the CIA and LGC datasets underwent pixel-level geometric correction. Each Landsat-MODIS (L-M) image pair was co-registered at a spatial resolution of 25 m, and pixelwise matching was performed by determining the optimal offset via maximization of the correlation function [42]. Consequently, the datasets inherently exhibit high spatial consistency. Under these conditions, additional global preprocessing may perturb the already corrected spectral relationships, thereby introducing extra errors and reducing prediction accuracy in the later stages.

B. Effectiveness of the Dual-Ranking Consistency Strategy

Similar-pixel selection incorporates representative reflectance information into the fine-resolution pixel prediction process, thereby enhancing prediction accuracy while preserving computational efficiency. During the OR stage, candidate pixels are confined to the same land-cover object. This constraint maintains consistency in land-cover composition and prevents the selection of pixels that exhibit similar temporal variations yet possess substantially different spectral characteristics. Under this condition, candidate pixels are required to exhibit only temporal change consistency, and

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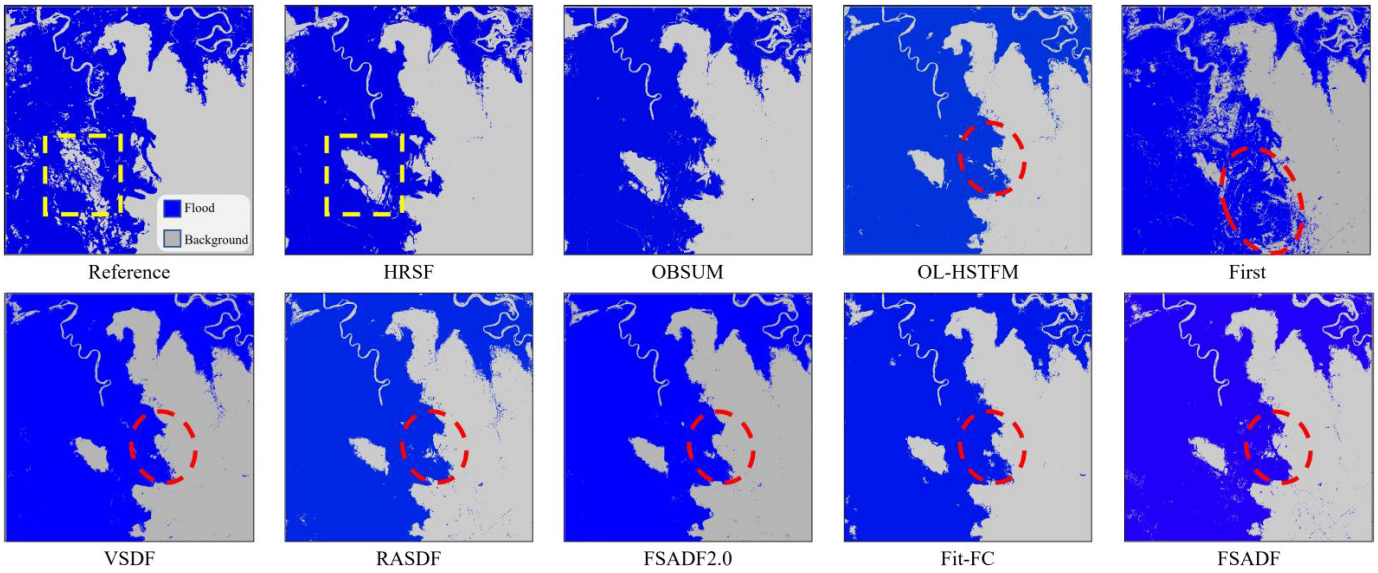


Fig. 10. Flood extent map derived from official true-color imagery and HRSF fusion imagery, February 9–13, 2019. The yellow-circled area illustrates inconsistent flood extent. Blue indicates affected areas.

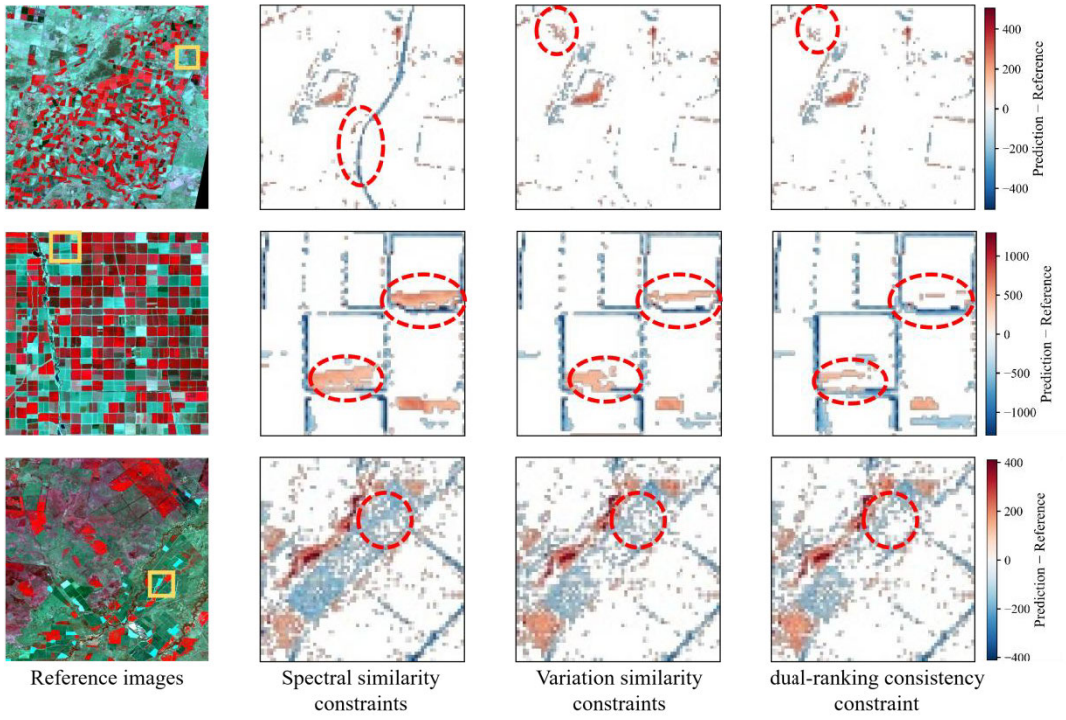


Fig. 11. Difference maps between the results of each strategy and the red band of the reference imagery. Red-circled areas correspond to the prediction outcomes of different strategies.

the residuals are allocated according to spatial distance-based weighting. To integrate neighborhood information from outside the object boundaries, the spatial constraint is relaxed in the CR stage, allowing candidate pixels to be selected beyond the original object. At this stage, candidate pixels are required to satisfy both spectral similarity and temporal change consistency. To meet this requirement, a dual-ranking consistency-based similar-pixel selection strategy is proposed. This strategy ensures that the selected pixels exhibit high consistency and representativeness according to multidimensional similarity criteria.

Comparative experiments were performed using three strategies: the spectral distance constraint, the temporal difference constraint, and the proposed dual-ranking consistency constraint. The results for the three study areas are shown in Fig. 11. The first column presents the Landsat imagery. The subsequent three columns illustrate the difference maps between each strategy and the red band of the reference image,

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TABLE VII
ACCURACY METRIC FOR THE RESULTS OF DIFFERENT STRATEGY COMBINATIONS

Site and date	Metric	Spectral similarity constraints	Variation similarity constraints	dual-ranking consistency constraint
CIA Feb. 12,2002	RMSE ↓	0.0726	0.0731	**0.0723**
	MAE ↓	**0.0503**	0.0512	0.0505
	r ↑	0.8750	0.8723	**0.8761**
	LBP ↓	0.0056	**0.0041**	0.0046
LGC Aug. 22,2004	RMSE ↓	0.0427	0.0428	**0.0426**
	MAE ↓	**0.0294**	0.0296	**0.0294**
	r ↑	0.9216	0.9215	**0.9219**
	LBP ↓	0.0037	0.0034	**0.0031**
IC site June 25,2023	RMSE ↓	0.0596	0.0593	**0.0585**
	MAE ↓	**0.0381**	0.0388	0.0382
	r ↑	0.9390	0.9394	**0.9411**
	LBP ↓	0.0074	0.0037	**0.0033**

corresponding to the magnified subregions highlighted by the yellow boxes in the first column.

In the red-circled areas of Fig. 11, the dual-ranking consistency constraint yields more accurate predictions for small objects. Under single-constraint strategies, small objects are more susceptible to influence from pixels of dominant land-cover types with larger spatial extents and differing change mechanisms. These pixels may dominate the similar-pixel selection process, thereby masking the true change signals of small objects during residual propagation. The dual-ranking consistency constraint requires candidate pixels to achieve high ranks in both similarity lists while maintaining a minimal ranking discrepancy. Pixels that are coincidentally similar in only one dimension, or those that are numerically dominant yet inconsistent in temporal change behavior, are thereby suppressed. Even when the number of available candidate pixels is small, pixels that accurately reflect the changing characteristics of small objects can still be retained for prediction. This mechanism enhances the semantic consistency of the similar-pixel set and improves prediction reliability in scenarios with complex land-cover change.

Table VII presents a quantitative comparison of fusion accuracy under different similar-pixel selection strategies across the three study areas. Overall, the fusion results obtained with the dual-ranking consistency constraint exhibit superior performance across all study areas, indicating that this strategy effectively enhances prediction accuracy during the similar-pixel selection stage. In the highly heterogeneous IC and CIA study areas, the dual-ranking consistency constraint yields lower RMSE values compared with the single-constraint strategies. Compared with the spectral distance constraint, RMSE is reduced by 0.0011 and 0.0003 in the IC and CIA study areas, respectively. Compared with the temporal difference constraint, RMSE decreases by 0.0008 in both study areas. In addition, in the IC study area, the dual-ranking consistency constraint reduces the LBP index by 0.0040 compared with the spectral distance constraint, further highlighting its advantage in preserving local structural characteristics. By contrast, in the LGC study area, where land-cover types are relatively homogeneous and temporal changes are moderate, the differences among the three similar-pixel selection strategies are minimal. This can primarily be attributed to the prevalence of large, homogeneous land-cover patches in this study area, where surface changes are relatively stable. Under such conditions, even a single-constraint strategy can successfully select representative similar pixels. The additional discriminative capability provided by the dual-ranking consistency constraint thus yields only marginal numerical improvement. Overall, the dual-ranking consistency constraint facilitates more reliable identification of semantically consistent similar pixels under complex surface conditions, thereby improving fusion prediction accuracy. In relatively homogeneous study areas, its advantage is primarily manifested in enhanced result stability rather than substantial numerical gains.

Furthermore, the RMSE decreases with increasing r_t in both sites while all other parameters are held constant as specified in Table I. This behavior can be attributed to the relaxation of the constraint, which permits the inclusion of a larger set of candidate pixels, whereas overly stringent constraints may result in insufficient information. As r_t further increases and exceeds the number of selected similar pixels, denoted as k_s, the RMSE gradually stabilizes across all sites. This behavior results from the CR step, in which higher rankings of V_k and S_k correspond to larger values of $Score_k$. After computing and ranking Score $_k$, the top k_s pixels are selected as similar pixels. Once r_t exceeds a certain threshold, the selection of similar pixels becomes invariant, and the RMSE consequently converges to a stable value. Therefore, r_t is generally set to be equal to k_s. However, the optimal value of r_t is dependent on the degree of surface homogeneity within the local window. Meanwhile, the CIA and IC sites are compared under similar change scenarios. Due to the more regular surface patterns and higher local homogeneity at the IC site, the ranking indices are more closely clustered, thereby requiring a relatively larger r_t to relax the constraint. As a result, the RMSE at the IC site continues to decrease within the examined range of r_t. Based on extensive experiments, the RMSE at the IC site stabilizes when r_t is approximately 70. In contrast, for the LGC site, characterized by higher homogeneity and smaller change magnitude, the RMSE stabilizes at approximately $r_t \approx 20$. In general, r_t can be set equal to k_s. For regions with larger changes and higher homogeneity, r_t can be moderately increased, while the resulting variation in RMSE remains within 0.001.

C. Statistical Significance and Robustness Analysis

To determine whether the observed improvements were consistent across prediction dates, Table VIII presents pairwise Wilcoxon signed-rank tests conducted using 26 prediction cases from the LGC and Mas datasets. HRSF significantly outperformed all eight comparison methods in terms of RMSE and $r(p \leq 5.91 \times 10^{-4})$, with win rates ranging from 73.1% to 100%. MAE produced a consistent result, with significant differences against all comparison methods ($p < 0.05$). These findings indicate that the spectral-temporal improvements of HRSF are systematic rather than being driven by a small number of favorable prediction dates. The LBP results reveal a clearer limitation. HRSF significantly outperformed VSDF,

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TABLE VIII
STATISTICAL COMPARISON BETWEEN HRSF AND COMPETING METHODS USING THE WILCOXON SIGNED-RANK TEST

Metric	Method	OBSUM	OL-HSTFM	VSDF	first	RASDF	FSDAF2.0	Fit-FC	FSDAF
RMSE	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	Wins	25	25	23	26	26	26	19	25
	Win rate(%)	96.15	96.15	88.46	100.00	100.00	100.00	73.08	96.15
	Mean improvement(%)	3.40	7.34	4.95	34.92	7.22	4.65	7.08	10.09
MAE	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.031	<0.001
	Wins	21	24	24	26	24	20	17	21
	Win rate(%)	80.77	92.31	92.31	100.00	92.31	76.92	65.38	80.77
	Mean improvement(%)	1.19	6.77	5.00	35.74	6.41	3.64	3.85	7.87
r	p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	Wins	25	25	22	25	26	26	19	24
	Win rate(%)	96.15	96.15	84.62	96.15	100.00	100.00	73.08	92.31
	Mean improvement(%)	0.44	1.62	0.65	18.57	1.36	1.08	1.92	2.82
LBP	p-value	0.727	0.105	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	Wins	14	11	26	26	26	26	25	26
	Win rate(%)	53.85	42.31	100.00	100.00	100.00	100.00	96.15	100.00
	Mean improvement(%)	-1.06	-20.42	63.18	77.82	49.41	51.05	39.40	53.99

First, RASDF, FSDAF2.0, Fit-FC, and FSDAF ($p < 10^{-6}$), but its differences from OBSUM ($p = 0.727$) and OL-HSTFM ($p = 0.105$) were not significant. This result is consistent with the performance observed for Mas, where the large ratio between coarse and fine resolutions increases the uncertainty introduced by the interpolated coarse image in Section II-F. Consequently, HRSF provides statistically robust improvements in spectral accuracy and temporal consistency, whereas its advantage in spatial-structure preservation remains dependent on the spatial-resolution relationship between the input observations.

VI. CONCLUSION

A novel HRSF is proposed in this study. By explicitly decomposing the residual transmission process into object-scale structural constraints and pixel-scale local adjustments, the proposed framework effectively alleviates the limitations of conventional approaches, which struggle to balance the preservation of spatial structure with the accuracy of temporal change reconstruction. Consequently, it achieves improved reconstruction performance for small land-cover features as well as newly emerged land-cover types within a given class. Furthermore, extensive experiments, including long-term time-series evaluation and flood mapping applications, verify that HRSF maintains reliable performance and exhibits strong robustness under varying temporal dynamics, sensor configurations, and practical application scenarios.

Despite its advantages, HRSF has several limitations. Due to the information constraints of coarse-resolution imagery, fine-scale details of small objects cannot be fully reconstructed, and reliance on a single fine-resolution reference limits the modeling of newly evolved structures. In cases of abrupt changes, using T_1 texture information may introduce structural artifacts. In addition, although HRSF shows robust performance across different landscapes, its applicability under more complex conditions, such as built-up areas with strong discontinuities and shadow effects, requires further investigation. Future work will incorporate multitemporal observations and heterogeneous auxiliary data to better capture dynamic changes and improve small-scale reconstruction [45], [46], [47]. Moreover, hybrid integration of the proposed residual-driven framework with deep learning models will be investigated to further improve the representation of complex spatiotemporal variations.

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